OFFICIAL TECHNICAL & WORKFLOW REPORT

Smart Campus Issue Reporting & Maintenance Platform

Complete Architecture, Authentication Workflow, Security Audit & Future Implementation Roadmap

Frontend Framework: React 19, TypeScript, Vite 8, Tailwind CSS v4, Lucide Icons	**Backend API:** Node.js, Express, MongoDB (Mongoose), JWT, BcryptJS
Security Protocol: 6-Digit SHA256 OTP, College Domain Enforcement, Server RBAC	**Security Audit Status:** **10/10 Verification Tests Passed** ■

1. End-to-End System Workflow

Step 1: Domain-Validated Registration

User submits details with an official college email address (e.g., @adhiyamaan.ac.in). Public self-registration automatically forces the STUDENT role on the backend.

Step 2: SHA-256 Hashed OTP Verification

Backend issues a 6-digit OTP (5-min expiry, 60s cooldown, max 3 attempts). Plaintext OTPs are never saved; only SHA-256 hashes are stored in the Otp model. Account creation is finalized only after OTP verification.

Step 3: Issue Reporting & Duplicate Check

Students/Faculty file complaints with title, location details, category, priority, and photo/video attachments. Real-time duplicate check prevents redundant tickets at the exact location.

Step 4: Admin Dispatch & Technician Assignment

Admins monitor master dispatch queue, review workload metrics, assign department technicians (Electrical, Plumbing, IT, Civil), and provision Faculty/Staff accounts.

Step 5: Repair Proof Upload & Resolution Feedback

Technicians accept tasks, update status to In Progress, and upload repair proof photos (afterImages). Reporter verifies repair completion and provides feedback ratings.

Step 6: Automated SLA Escalation Engine

Periodic background service monitors SLA target dates every 10 minutes. Overdue complaints automatically trigger SLA breach escalation flags (isEscalated: true).

2. System Security & Role-Based Access Control (RBAC)

Every protected route queries MongoDB directly in authMiddleware.ts using the token's userId. This ensures role claims submitted by clients or modified in local browser state are strictly ignored.

User Role	Creation Method	Permitted System Capabilities
Student	Public OTP Registration	File issue tickets, view own reported complaints, upvote campus issues, track ticket status, verify completed repairs, submit feedback.
Faculty	Admin Provisioned	Raise department tickets, view campus complaints, track status, verify completed maintenance.
Technician Staff	Admin Provisioned	View department queue, accept assigned tasks, update status (In Progress / Resolved), upload repair proof photos.
Admin	Seeded / Super Admin	Master dispatch queue, technician assignment, user management (activate/deactivate), privileged account creation, audit logs, CSV export.

3. Security Audit & Test Results

#	Security Scenario Tested	Expected & Observed Security Behavior	Status
1	**Gmail Registration Attempt**	Rejected with 400 ('Please use official college email')	**PASS** ■
2	**Valid Institutional Email Registration**	Accepted; 6-digit OTP code generated & printed	**PASS** ■
3	**Incorrect OTP Submission**	Rejected with 400 ('Invalid OTP. Please try again')	**PASS** ■
4	**Expired / Invalid OTP Attempt**	Rejected with 400 ('OTP has expired')	**PASS** ■
5	**Student Accessing Admin API**	Rejected with 403 Forbidden	**PASS** ■
6	**Client Role Spoofing (Student ➔ Admin)**	Backend queries DB role and rejects with 403	**PASS** ■
7	**Self-Registration as Admin Role**	Public endpoint forces STUDENT role & blocks Admin	**PASS** ■
8	**Technician Accessing Admin API**	Rejected with 403 Forbidden	**PASS** ■
9	**Unverified User Login Attempt**	Blocked with 403 requiring OTP email verification	**PASS** ■
10	**Admin Provisioning Faculty Account**	Successfully created with unverified state (201)	**PASS** ■

4. Recommended Next Phase Features

1. Live SMTP Email & SMS Gateway	Integrate Nodemailer (SendGrid / AWS SES) for live OTP dispatch and Twilio for SMS alerts on ticket updates.
2. Real-Time WebSockets (Socket.io)	Instant push notifications for technician task assignments and live ticket status timeline updates.
3. QR Code Room Scan Reporting	Printable QR codes placed on classroom/lab doors to automatically pre-fill location details on mobile scanners.
4. Spatial Heatmap Analytics	Interactive 2D campus map visually highlighting high-maintenance building zones in real-time.
5. AI Vision Photo Categorization	Machine learning model to analyze uploaded damage photos and auto-suggest fault category and priority.